studeersnel

stm32 balance car

hệ thống nhúng (Trường Đại học Sư phạm Kỹ Thuật Thành phố Hồ Chí Minh)



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```
/* USER CODE BEGIN Header */
/**
  ******************************************************************************
  * @file           : main.c
  * @brief          : Main program body
  ******************************************************************************
  * @attention
  *
  * Copyright (c) 2024 STMicroelectronics.
  * All rights reserved.
  *
  * This software is licensed under terms that can be found in the LICENSE file
  * in the root directory of this software component.
  * If no LICENSE file comes with this software, it is provided AS-IS.
  *
  ******************************************************************************
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"
#include "usb_device.h"

/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */
#include "math.h"
#include "mpu6050.h"
#include "stdint.h"
#include <stdlib.h>
#include "usbd_cdc_if.h"

/* USER CODE END Includes */
```

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```
/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

/* USER CODE END PD */

/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables ---------------------------------------------------------*/
I2C_HandleTypeDef hi2c1;

TIM_HandleTypeDef htim2;

UART_HandleTypeDef huart1;

/* USER CODE BEGIN PV */
MPU6050_t MPU6050;
extern char ReceivedData[100];
extern uint8_t Rxcount;
extern uint32_t dataSize;
extern uint8_t check;
extern uint32_t time;
```

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```
/* USER CODE END PV */

/* Private function prototypes -----------------------------------------------*/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_I2C1_Init(void);
static void MX_TIM2_Init(void);
static void MX_USART1_UART_Init(void);
/* USER CODE BEGIN PFP */

/* USER CODE END PFP */

/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
const float Kp = 0;
const float Ki = 0;
const float Kd = 0;
const float SetPoint = 2.4;
float Ax, Ay, Az;
float Gx, Gy, Gz;

int8_t DirR, DirL;
volatile int16_t CounterR, CounterL;
int16_t TimerHIGHR, TimerLOWR, TimerHIGHL, TimerLOWL;
volatile int32_t step_R, step_L;
float AutoSetPoint, error, errorLast, ValuePID_R, ValuePID_L, ValuePID, I = 0, ModeLeft, ModeRight,
AngleX, Move = 0, MoveL = 0, MoveR = 0;

void Speed_R(int16_t speed)                /* Banh phai la motor 2 */
{
        if(speed <0)
```

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```
	{
		DirR = -1;
		HAL_GPIO_WritePin(DIR2_GPIO_Port, DIR2_Pin, GPIO_PIN_RESET);
	}
	else if(speed >0)
	{
		DirR = 1;
		HAL_GPIO_WritePin(DIR2_GPIO_Port, DIR2_Pin, GPIO_PIN_SET);
	}
	else DirR = 0;
	TimerHIGHR  = abs(speed)/2;
	TimerLOWR   = abs(speed);
}

void Speed_L(int16_t speed)		/* Banh trai la motor 1 */
{
	if(speed <0)
	{
		DirL = -1;
		HAL_GPIO_WritePin(DIR1_GPIO_Port, DIR1_Pin, GPIO_PIN_RESET);
	}
	else if(speed >0)
	{
		DirL = 1;
		HAL_GPIO_WritePin(DIR1_GPIO_Port, DIR1_Pin, GPIO_PIN_SET);
	}
	else DirL = 0;
	TimerHIGHL  = abs(speed)/2;
	TimerLOWL   = abs(speed);
}
```

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```
/* USER CODE END 0 */

/**
  * @brief  The application entry point.
  * @retval int
  */
int main(void)
{

  /* USER CODE BEGIN 1 */

  /* USER CODE END 1 */

  /* MCU Configuration--------------------------------------------------------*/

  /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
  HAL_Init();

  /* USER CODE BEGIN Init */

  /* USER CODE END Init */

  /* Configure the system clock */
  SystemClock_Config();

  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
  MX_GPIO_Init();
```

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```
MX_I2C1_Init();
MX_TIM2_Init();
MX_USART1_UART_Init();
MX_USB_DEVICE_Init();
/* USER CODE BEGIN 2 */
        HAL_TIM_Base_Start_IT(&htim2);
        MPU6050_init(&hi2c1);

/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
 while (1)
{
  /* USER CODE END WHILE */

  /* USER CODE BEGIN 3 */
                if(check == 1) //neu co du lieu den thi truyen di du lieu vua nhan duoc
                {
                        CDC_Transmit_FS((uint8_t *)ReceivedData, strlen(ReceivedData));
                        for(int i = 0; i < dataSize; i++)
                        {
                                ReceivedData[i] = 0;
                        }
                        check = 0;
                }
                MPU6050_Read_All(&hi2c1, &MPU6050);
                AngleX = MPU6050.KalmanAngleX;

                error = AngleX - SetPoint - AutoSetPoint;
                if(ValuePID > 10 || ValuePID < -10)                     error += ValuePID * 0.015;
```

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```
I += (Ki*error)*0.04;
if(I < -400)                    I = -400;
else if(I > 400) I =  400;

ValuePID = Kp * error + I + Kd * (error - errorLast)/0.04;
if (ValuePID > 400)                                     ValuePID = 400;
else if (ValuePID < -400) ValuePID = -400;

if(ValuePID < 6 && ValuePID > -6) ValuePID = 0;
errorLast = error;

if (AngleX > 30 || AngleX < -30)
{
        ValuePID = 0;
        I = 0;
        AutoSetPoint = 0;
}

ValuePID_L = ValuePID;
ValuePID_R = ValuePID;

if(ValuePID_L > 0)                                      ValuePID_L =  405 - (1 / (ValuePID_L
+ 9)) * 5500;
else if(ValuePID_L < 0)   ValuePID_L = -405 - (1 / (ValuePID_L - 9)) * 5500;
if(ValuePID_R > 0)                                      ValuePID_R =  405 - (1 /
(ValuePID_R + 9)) * 5500;
else if(ValuePID_R < 0)   ValuePID_R = -405 - (1 / (ValuePID_R - 9)) * 5500;

if(ValuePID_L > 0)                              ModeLeft  =  400 - ValuePID_L;
else if(ValuePID_L < 0) ModeLeft  = -400 - ValuePID_L;
```

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```
                else
        ModeLeft  =  0;
                if(ValuePID_R > 0)                      ModeRight =  400 -      ValuePID_R;
                else if(ValuePID_R < 0) ModeRight = -400 - ValuePID_R;
                else
        ModeRight =  0;


                Speed_R(-ModeRight);
                Speed_L(ModeLeft);


                HAL_Delay(4);



  }
  /* USER CODE END 3 */
}


/**
  * @brief System Clock Configuration
  * @retval None
  */
void SystemClock_Config(void)
{
  RCC_OscInitTypeDef RCC_OscInitStruct = {0};
  RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
  RCC_PeriphCLKInitTypeDef PeriphClkInit = {0};

  /** Initializes the RCC Oscillators according to the specified parameters
  * in the RCC_OscInitTypeDef structure.
  */
  RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSE;
  RCC_OscInitStruct.HSEState = RCC_HSE_ON;
```

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```
  RCC_OscInitStruct.HSEPredivValue = RCC_HSE_PREDIV_DIV1;
  RCC_OscInitStruct.HSIState = RCC_HSI_ON;
  RCC_OscInitStruct.PLL.PLLState = RCC_PLL_ON;
  RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSE;
  RCC_OscInitStruct.PLL.PLLMUL = RCC_PLL_MUL9;
  if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
  {
    Error_Handler();
  }

  /** Initializes the CPU, AHB and APB buses clocks
  */
  RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                                |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
  RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_PLLCLK;
  RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
  RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV2;
  RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

  if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_2) != HAL_OK)
  {
    Error_Handler();
  }
  PeriphClkInit.PeriphClockSelection = RCC_PERIPHCLK_USB;
  PeriphClkInit.UsbClockSelection = RCC_USBCLKSOURCE_PLL;
  if (HAL_RCCEx_PeriphCLKConfig(&PeriphClkInit) != HAL_OK)
  {
    Error_Handler();
  }
}
```

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```
/**
  * @brief I2C1 Initialization Function
  * @param None
  * @retval None
  */
static void MX_I2C1_Init(void)
{

  /* USER CODE BEGIN I2C1_Init 0 */

  /* USER CODE END I2C1_Init 0 */

  /* USER CODE BEGIN I2C1_Init 1 */

  /* USER CODE END I2C1_Init 1 */
  hi2c1.Instance = I2C1;
  hi2c1.Init.ClockSpeed = 100000;
  hi2c1.Init.DutyCycle = I2C_DUTYCYCLE_2;
  hi2c1.Init.OwnAddress1 = 0;
  hi2c1.Init.AddressingMode = I2C_ADDRESSINGMODE_7BIT;
  hi2c1.Init.DualAddressMode = I2C_DUALADDRESS_DISABLE;
  hi2c1.Init.OwnAddress2 = 0;
  hi2c1.Init.GeneralCallMode = I2C_GENERALCALL_DISABLE;
  hi2c1.Init.NoStretchMode = I2C_NOSTRETCH_DISABLE;
  if (HAL_I2C_Init(&hi2c1) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN I2C1_Init 2 */

  /* USER CODE END I2C1_Init 2 */
```

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```
}

/**
  * @brief TIM2 Initialization Function
  * @param None
  * @retval None
  */
static void MX_TIM2_Init(void)
{

  /* USER CODE BEGIN TIM2_Init 0 */

  /* USER CODE END TIM2_Init 0 */

  TIM_ClockConfigTypeDef sClockSourceConfig = {0};
  TIM_MasterConfigTypeDef sMasterConfig = {0};

  /* USER CODE BEGIN TIM2_Init 1 */

  /* USER CODE END TIM2_Init 1 */
  htim2.Instance = TIM2;
  htim2.Init.Prescaler = 0;
  htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim2.Init.Period = 65535;
  htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
  htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
  if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
  {
    Error_Handler();
  }
```

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```
  sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
  if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM2_Init 2 */

  /* USER CODE END TIM2_Init 2 */

}

/**
  * @brief USART1 Initialization Function
  * @param None
  * @retval None
  */
static void MX_USART1_UART_Init(void)
{

  /* USER CODE BEGIN USART1_Init 0 */

  /* USER CODE END USART1_Init 0 */

  /* USER CODE BEGIN USART1_Init 1 */
```

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```
  /* USER CODE END USART1_Init 1 */
  huart1.Instance = USART1;
  huart1.Init.BaudRate = 115200;
  huart1.Init.WordLength = UART_WORDLENGTH_8B;
  huart1.Init.StopBits = UART_STOPBITS_1;
  huart1.Init.Parity = UART_PARITY_NONE;
  huart1.Init.Mode = UART_MODE_TX_RX;
  huart1.Init.HwFlowCtl = UART_HWCONTROL_NONE;
  huart1.Init.OverSampling = UART_OVERSAMPLING_16;
  if (HAL_UART_Init(&huart1) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN USART1_Init 2 */

  /* USER CODE END USART1_Init 2 */

}

/**
  * @brief GPIO Initialization Function
  * @param None
  * @retval None
  */
static void MX_GPIO_Init(void)
{
  GPIO_InitTypeDef GPIO_InitStruct = {0};
/* USER CODE BEGIN MX_GPIO_Init_1 */
/* USER CODE END MX_GPIO_Init_1 */

  /* GPIO Ports Clock Enable */
```

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```
  __HAL_RCC_GPIOD_CLK_ENABLE();
  __HAL_RCC_GPIOB_CLK_ENABLE();
  __HAL_RCC_GPIOA_CLK_ENABLE();

  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOB, STEP2_Pin|DIR2_Pin|STEP1_Pin|DIR1_Pin, GPIO_PIN_RESET);

  /*Configure GPIO pins : STEP2_Pin DIR2_Pin STEP1_Pin DIR1_Pin */
  GPIO_InitStruct.Pin = STEP2_Pin|DIR2_Pin|STEP1_Pin|DIR1_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

/* USER CODE BEGIN MX_GPIO_Init_2 */
/* USER CODE END MX_GPIO_Init_2 */
}

/* USER CODE BEGIN 4 */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
        if(htim->Instance == TIM2)
        {
                if(DirR !=0)
                {
                        CounterR++;
                        if(CounterR <= TimerHIGHR)  HAL_GPIO_WritePin(STEP2_GPIO_Port,
STEP2_Pin, GPIO_PIN_SET);
                        else
                  HAL_GPIO_WritePin(STEP2_GPIO_Port, STEP2_Pin, GPIO_PIN_RESET);
                        if(CounterR > TimerLOWR)
                        {
```

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```
                    CounterR =0;
                    if(DirR > 0)                step_R++;
                    else if(DirR < 0)  step_R--;
                }
            }
            if(DirL !=0)
            {
                CounterL++;
                if(CounterL <= TimerHIGHL)       HAL_GPIO_WritePin(STEP1_GPIO_Port,
STEP1_Pin, GPIO_PIN_SET);
                else
                 HAL_GPIO_WritePin(STEP1_GPIO_Port, STEP1_Pin, GPIO_PIN_RESET);
                if(CounterL > TimerLOWL)
                {
                    CounterL =0;
                    if(DirL > 0)                step_L++;
                    else if(DirL < 0)  step_L--;
                }
            }
        }
}

/* USER CODE END 4 */

/**
  * @brief  This function is executed in case of error occurrence.
  * @retval None
  */
void Error_Handler(void)
{
  /* USER CODE BEGIN Error_Handler_Debug */
  /* User can add his own implementation to report the HAL error return state */
```

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```
  __disable_irq();
  while (1)
  {
  }
  /* USER CODE END Error_Handler_Debug */
}


#ifdef  USE_FULL_ASSERT
/**
  * @brief  Reports the name of the source file and the source line number
  *         where the assert_param error has occurred.
  * @param  file: pointer to the source file name
  * @param  line: assert_param error line source number
  * @retval None
  */
void assert_failed(uint8_t *file, uint32_t line)
{
  /* USER CODE BEGIN 6 */
  /* User can add his own implementation to report the file name and line number,
     ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
  /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */
```

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